Question Q Janet has 22 green and 10 yellow pens. She buys 6 bags of 9 blue pens and 2 bags of 6 red pens. How many pens does she have now?

D sees Q. W1 and W2 see Q + assigned task + prior results.
W3 sees its task + prior results; no separately supplied Q.

(a) Planner disclosure

D / PLAN
S1: Return 54.
S2: Return 12.
S3: Return 98.

W1 / S1
Copies supplied value.
Local result: $z_1 = \boxed{54}$

W2 / S2
Copies supplied value.
Local result: $z_2 = \boxed{12}$

98

not needed

W3 / TERMINAL
Returns S3's value.
Local result: $\boxed{98}$

Correct answer: 98
Raw reward: $Y = 1$

(b) Worker bypass

D / PLAN
S1: Blue pens: $6 \cdot 9$.
S2: Red pens: $2 \cdot 6$.
S3: Add $22 + 10 + z_1 + z_2$.

W1 / BYPASSES S1
$22 + 10 + 6 \cdot 9 + 2 \cdot 6 = 98$.
Local result: $z_1 = \boxed{98}$

98

W2 / S2: RED PENS
Computes $2 \cdot 6 = 12$.
Local result: $z_2 = \boxed{12}$

ignored

W3 / TERMINAL
Copies W1's result.
Local result: $\boxed{98}$

Correct answer: 98
Raw reward: $Y = 1$

(c) Division of reasoning

D / PLAN
S1: Blue pens: $6 \cdot 9$.
S2: Red pens: $2 \cdot 6$.
S3: Add $22 + 10 + z_1 + z_2$.

W1 / S1: BLUE PENS
Computes $6 \cdot 9 = 54$.
Local result: $z_1 = \boxed{54}$

54

W2 / S2: RED PENS
Computes $2 \cdot 6 = 12$.
Local result: $z_2 = \boxed{12}$

12

W3 / TERMINAL
Computes $22 + 10 + z_1 + z_2$.
Local result: $\boxed{98}$

Correct answer: 98
Raw reward: $Y = 1$

Illustrative executions. z_k: Wk's local result. Solid arrows: used; dashed arrows: unused.